

TWO-STAGE MESSAGE-BOUND GATE

$m = \text{merge}(A, B)$: r_A (7d) and r_B (30d) are both genuine ancestors

STAGE 1 · PER-RECORD GATE

authorize each independently · 8 conjunctive checks (supp.)

write-event causal precedence

integrity flow (Biba): $l(\text{rec}) \geq l(m)$

$r_A \rightarrow \text{ADMIT}$ $r_B \rightarrow \text{ADMIT}$

admitted set $\{ r_A, r_B \}$

STAGE 2 · SET-LEVEL VIEW

governs the served set (same logical_key K)

no cert → **REQUIRE_RESOLUTION**

unresolved union denied; never served jointly

signed cert → **serve adopted only**

(cert bound to the exact conflict set)

Reachable $\neq$ adopted: per-record admits both;
the set-level gate denies the unresolved union.